

Machine Learning Assignment 1: Polynomial Regression

Abhinav Bhatia
BT2024156

GitHub Repository:

<https://github.com/abhinavbhatia2006/machine-learning-polynomial-regression>

1 Objective

The assignment requires polynomial regression models to predict a continuous target variable y from the supplied input variables. The two personalized problems are treated as separate datasets. Part 1 concerns the Net Power Score of a geothermal power plant and allows polynomial degrees up to 10. Part 2 concerns a Thermal Anomaly Score over a 3D geological region and allows polynomial degrees up to 20. The assignment specifies Mean Squared Error (MSE) and R^2 as the main evaluation metrics and asks for the selection of an appropriate polynomial degree while avoiding underfitting and overfitting.

The implementation uses Python with NumPy, Pandas and scikit-learn. The complete training process is contained in `p1.py` and `p2.py`, while the prediction scripts use the selected configurations to generate the required submission files.

2 Overall Methodology

For each problem, the same modelling workflow is followed. The training data are loaded and the required feature columns are separated from the target. A polynomial feature transformation is then applied for every degree in the allowed range. For each degree, four regression approaches are compared:

1. Ordinary Least Squares (OLS)
2. Ridge Regression (L2 regularization)
3. Lasso Regression (L1 regularization)
4. Elastic Net Regression (L1 + L2 regularization)

Every candidate model is evaluated using shuffled 5-fold cross-validation with `random_state=42`. In each fold, the model is trained on 80% of the data and evaluated on the remaining 20%. The five validation MSE values are averaged to obtain the model's validation MSE. The same procedure is used to calculate the average training MSE and the corresponding training and validation R^2 values.

The final model is selected using only the **lowest average validation MSE**. Train MSE and R^2 are reported as additional diagnostics, but they do not affect model selection.

3 Feature Scaling and Polynomial Features

Feature scaling is used because polynomial expansion can create terms with very different numerical magnitudes. For example, a feature with a relatively large range can produce much larger squared, cubic, or higher-order values than another feature. This can make numerical optimization poorly conditioned and can also make regularization uneven because L1 and L2 penalties operate directly on coefficient magnitudes.

The implemented pipeline therefore uses two scaling stages:

Input features $\rightarrow$ StandardScaler $\rightarrow$ PolynomialFeatures $\rightarrow$ StandardScaler $\rightarrow$ Regression model.

The first scaler standardizes the original input variables before polynomial expansion. The second scaler standardizes the resulting polynomial terms. Both scalers are inside the scikit-learn pipeline, so their parameters

are learned from the training portion of each cross-validation fold rather than from the complete dataset. This prevents preprocessing information from leaking from the validation fold into the training fold.

The polynomial transformation uses `include_bias=False`, since the regression models already handle the intercept. A polynomial degree d includes monomials whose total feature-power sum is at most d , including interaction terms.

4 Choice of Regression Models

4.1 Ordinary Least Squares

OLS is included as the unregularized baseline. It allows the experiment to determine whether adding regularization actually improves validation performance. Without this baseline, it would not be possible to separate the benefit of polynomial expansion from the benefit of L1 or L2 penalties.

4.2 Ridge Regression (L2)

Ridge is useful because high-degree polynomial features are strongly correlated with one another. The L2 penalty discourages excessively large coefficients and generally provides a more stable fit when many correlated polynomial terms are present. It is therefore a natural model for controlling variance without forcing coefficients exactly to zero.

4.3 Lasso Regression (L1)

Lasso adds an L1 penalty and can shrink some polynomial coefficients exactly to zero. This can reduce the effective complexity of a large polynomial model and can help when many generated terms are unnecessary. Lasso was therefore included to test whether a sparse polynomial representation generalizes better than an unregularized or purely L2-regularized model.

4.4 Elastic Net

Elastic Net combines L1 and L2 regularization. It was included because polynomial terms are highly correlated, which is a setting where a pure L1 penalty can behave differently from L2 regularization. Elastic Net provides a compromise: it can encourage sparsity through L1 while retaining the stabilizing effect of L2.

5 Choice of Alpha and L1 Ratio

For Ridge, Lasso and Elastic Net, the code tests the following alpha values:

$$\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}.$$

These values were chosen to cover a wide range of regularization strengths, from 10^{-3} to 10^2 , on a logarithmic scale. The purpose is to include both very weak regularization, where the model behaves more like ordinary polynomial regression, and strong regularization, where coefficients are heavily constrained. A logarithmic progression is appropriate because the meaningful change in regularization strength is usually multiplicative rather than additive. Six values also provide a practical balance between search coverage and computational cost.

For Elastic Net, the following L1 ratios are tested:

$$\text{l1_ratio} \in \{0.1, 0.25, 0.5, 0.75, 0.9\}.$$

An L1 ratio close to 0 gives a model dominated by L2 regularization, while a value close to 1 gives a model dominated by L1 regularization. The selected values therefore cover the range from mostly L2 through balanced L1/L2 to mostly L1. The exact endpoints 0 and 1 are not needed because pure L2 and pure L1 are already represented separately by Ridge and Lasso.

6 Random Coordinate Selection in Lasso and Elastic Net

Lasso and Elastic Net are solved using iterative coordinate-descent optimization. The polynomial feature expansion produces many terms, and at the higher degrees the feature matrix contains a large number of strongly correlated columns. This can make convergence slower, especially for small values of alpha where regularization is weak.

The implementation therefore uses `selection="random"` for Lasso and Elastic Net. Random coordinate selection changes the order in which coefficient coordinates are updated. In high-dimensional and correlated problems, this can allow the optimization to reach the convergence criterion more efficiently than a fixed cyclic order. It is used to improve practical convergence rather than to change which original input variables are considered. All supplied input features are still used in the polynomial expansion.

A fixed `random_state=42` is used so that the random coordinate order is reproducible across runs. The models also use a large maximum iteration limit of 200,000 and a convergence tolerance of 10^{-4} to reduce the chance of premature termination.

7 Cross-Validation and Model Selection

A shuffled 5-fold split is used for every candidate model. The same fixed splitting strategy makes the comparison between models more consistent. For a model, the average validation MSE is

$$\text{MSE}_{val} = \frac{1}{5} \sum_{k=1}^5 \text{MSE}_k,$$

where each MSE_k is calculated only on the held-out validation fold.

The model-selection rule is intentionally simple:

$$\text{Best model} = \arg \min \text{MSE}_{val}.$$

Validation MSE is used because the objective is prediction on unseen test data. Training MSE measures how well a model fits data it has already seen and can become artificially small for high-degree models. Therefore, training MSE is reported only to understand model behavior, while the validation MSE is the sole selection criterion.

The coefficient of determination is also reported:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

A larger R^2 indicates that more of the target variance is explained by the model. The assignment lists both MSE and R^2 as evaluation metrics.

8 Parallel Model Evaluation

The many model configurations are evaluated using `ThreadPoolExecutor` with five worker threads. This allows up to five independent model configurations to be processed concurrently. Inside each individual cross-validation operation, `n_jobs=1` is used so that parallelism is controlled by the outer five-thread executor rather than creating additional nested threads.

8.1 Part 1 Search Size

Part 1 tests degrees 1 through 10. The number of configurations is:

- OLS: 10
- Ridge: $10 \times 6 = 60$
- Lasso: $10 \times 6 = 60$

- Elastic Net: $10 \times 6 \times 5 = 300$

Therefore, a total of **430 models** are evaluated.

8.2 Part 2 Search Size

Part 2 tests degrees 1 through 20. The number of configurations is:

- OLS: 20
- Ridge: $20 \times 6 = 120$
- Lasso: $20 \times 6 = 120$
- Elastic Net: $20 \times 6 \times 5 = 600$

Therefore, a total of **860 models** are evaluated.

9 Results

9.1 Part 1: Power Plant Steam Turbine Optimization

The best configuration was a degree-5 Lasso model with $\alpha = 0.01$.

Metric	Result
Models tested	430
Best polynomial degree	5
Best model	Lasso (L1)
Alpha	0.01
Average Train MSE	0.2271912594
Average Validation MSE	0.3037583528
Average Train R^2	0.9779475432
Average Validation R^2	0.9701815630

The degree-5 model produced the lowest validation MSE among all 430 tested configurations. The corresponding validation R^2 of approximately 0.9702 indicates that the selected polynomial model captures most of the variation in the target while achieving the best cross-validated prediction error among the candidates.

9.2 Part 2: Subterranean Thermal Reservoir Mapping

The best configuration was a degree-11 Elastic Net model with $\alpha = 0.001$ and L1 ratio = 0.1.

Metric	Result
Models tested	860
Best polynomial degree	11
Best model	Elastic Net
Alpha	0.001
L1 ratio	0.1
Average Train MSE	0.1655382392
Average Validation MSE	0.2644830804
Average Train R^2	0.9964517668
Average Validation R^2	0.9942217064

The degree-11 Elastic Net configuration produced the lowest validation MSE among the 860 candidates. Its validation R^2 of approximately 0.9942 indicates a very strong fit to the observed thermal-anomaly relationship while retaining good cross-validated predictive performance.

10 Final Training and Inference

After model selection, the chosen configuration for each problem is fitted on the complete corresponding training dataset. The preprocessing steps and selected regression model remain in the same pipeline used during model evaluation.

For Part 1, the final inference model is:

Degree 5 polynomial + Lasso($\alpha = 0.01$).

For Part 2, the final inference model is:

Degree 11 polynomial + Elastic Net($\alpha = 0.001$, $\text{l1_ratio} = 0.1$).

The prediction scripts `p1_predict.py` and `p2_predict.py` load the corresponding test data, apply the trained pipeline, and write the predictions to:

- `BT2024156_pred_var1.csv`
- `BT2024156_pred_var2.csv`

Each prediction file contains a single column named `y`, matching the required submission format.

11 Conclusion

The assignment was approached as a systematic polynomial-regression model-selection problem. Polynomial degrees were searched over the complete ranges permitted by the two problems, and OLS, Ridge, Lasso and Elastic Net were compared using shuffled 5-fold cross-validation. Feature scaling was used to improve numerical conditioning and to make regularization meaningful across polynomial terms. The alpha grid was selected to span weak to strong regularization on a logarithmic scale, while the Elastic Net L1 ratios covered mostly-L2, balanced and mostly-L1 mixtures. Random coordinate selection was used for Lasso and Elastic Net to improve convergence in the high-dimensional, highly correlated polynomial feature space, with a fixed random seed for reproducibility.

The final decisions were made solely using the lowest average validation MSE. Part 1 selected degree 5 Lasso with $\alpha = 0.01$, while Part 2 selected degree 11 Elastic Net with $\alpha = 0.001$ and L1 ratio 0.1. The resulting models are retrained on the complete training data and used to generate the final test predictions.